Olle Marmenlind

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Final-year Computer Engineering student at KTH with internships in embedded and full-stack engineering and experience as Technical Product Owner at Sevan AB. Primary interests include backend and systems engineering. Available for a software engineering internship or thesis project from early 2027, ahead of graduation in June 2027.

Education

Royal Institute of Technology (KTH) Stockholm, Sweden
B.Sc. in Computer Engineering **Expected June 2027**

- **Relevant Coursework:** Data Structures & Algorithms, Distributed Systems, Operating Systems, Networks, Databases; graduate-level Embedded Systems and Network Security
- **Certifications:** CCNA, CCNP

Engineering Experience

Motillo AB Stockholm, Sweden
Software Engineer Intern **March 2026 – June 2026**

- Reduced query times from 330 ms to ∼20 ms over 4 million rows by combining PostgreSQL materialized views for historical rollups with live data.
- Built a React/TypeScript control interface for UptimeRobot monitors and Litium order ingestion with Entra ID authentication and support for legacy Chrome 76 kiosks.
- Built an API for Elasticsearch order data with asynchronous ingestion using Hangfire and Polly to handle upstream rate limits, and OpenTelemetry for observability.

Proximion AB Stockholm, Sweden
R&D Systems Engineer Intern **October 2025 – January 2026**

- Built a multithreaded FreeRTOS pipeline with non-blocking IPC and an MCU-to-PC binary protocol using state-machine framing, sequence tracking, and CRC validation.
- Wrote kernel-level hooks to detect stack overflows and heap exhaustion, extracting thread-specific crash states through out-of-band hardware signaling without a debugger.
- Configured cross-platform GNU Make builds and self-hosted Linux CI/CD using hermetic Docker containers for reproducible builds and automated releases.

Other Professional Experience

Sevan AB Stockholm, Sweden (Remote)
Technical Product Owner **March 2022 – Present**

- Promoted to Product Owner, overseeing B2C platform growth from 4.5K to 200K users and migration of 300+ B2B accounts to e-commerce.
- Directed the roadmap, specified API schemas and data models, and coordinated development across external teams and vendors.

Selected Projects — continued on page 2

- **Token Catalog:** Built a Go pipeline and live web service providing versioned, verified Material Design token data for developers and coding agents.
- **Linduistics (WIP):** Developing a lexical analysis platform combining articulatory phonetic comparison with multilingual word-sense alignment.

Technical Skills

Languages: Java, C++, Go, Python, TypeScript, C#
Systems & Tools: Linux, FreeRTOS, Docker, CI/CD, GNU Make, Git, TCP/UDP, OpenTelemetry
Web & Data: React, HTMX, Spring Boot, ASP.NET Core, REST, PostgreSQL, Elasticsearch, ETL

Selected Projects — continued

Linduistics Work in progress

Stack: Python, Go, PostgreSQL, React, TypeScript, spaCy, sentence-transformers

Developing a full-stack application for comparing vocabulary, pronunciation, and meaning across five Germanic languages, with a comparative lexicon, text inspector, and language games.

- Implemented weighted edit distance over articulatory features using a precomputed sound-distance matrix, Numba JIT compilation, and rolling buffers that keep working memory linear in pronunciation length.
- Built a dataset pipeline that aligns individual word senses through translation-graph mining, spelling collisions, and multilingual embeddings while preserving source identity, etymological evidence, and unresolved ambiguity.
- Implemented reproducible, resumable builds with field-level regression checks and consistent numerical behavior across Python, Parquet, and PostgreSQL, publishing comparisons and lookup tables in one transaction.
- Built a React/TypeScript frontend backed by Go and Python NLP services, with an interactive 2D/3D language map using multidimensional scaling and iterative stress minimization.

Token Catalog tokencatalog.dev

github.com/sojupie/tokens

Stack: Go, SQLite, Chromium/chromedp, templ, HTMX, Docker

Built and deployed a service to make Material Design 3 token values directly usable in custom web components and agent-assisted development, after encountering friction accessing the data while styling a React application.

- Applied extracted tokens to the catalog's templ/HTMX interface, generating colors, typography, and shapes across six light/dark and contrast combinations, and exposed the data through context-filtered JSON exports and read-only MCP tools.
- Built a concurrent Go crawler using Chromium's DevTools Protocol to capture dynamically loaded JSON payloads, tracking network activity and request completion with cached captures and targeted retries.
- Implemented semantic change detection for token values and dependency trees across platform and theme contexts, normalizing unstable upstream IDs and ordering to avoid reporting metadata churn as design changes.
- Designed a release workflow with immutable snapshots, completeness checks, and reviews bound to artifact hashes, with stale-candidate rejection, atomic publication, and recovery from interrupted activation.
- Reduced verification memory overhead through incremental JSON decoding and per-record schema validation, and loaded published SQLite data into memory at startup so token reads require no database queries or upstream access.
- Achieved 100/100 in performance, accessibility, best practices, and SEO in a mobile PageSpeed Insights lab audit (11 October 2026). Lighthouse 13.5 emulated a Moto G Power with simulated Slow 4G (150 ms RTT, 1.64 Mb/s throughput) from Europe: 1.2 s largest contentful paint, 0 ms total blocking time, and zero cumulative layout shift.
- Implemented OpenAPI, Markdown responses, and MCP discovery, passing all 4 PageSpeed agentic browsing checks. Cloudflare's Is It Agent Ready? audit awarded 100/100 for discoverability, content, and bot access control. The overall score was 80/100 because the public, read-only service omits authentication and registration.